

Git HOL - Branching and Merging

Create New Branch

Command:

```
git branch GitNewBranch
```

Terminal Output:

```
$ git branch GitNewBranch
```

List Local and Remote Branches

Command:

```
git branch -a
```

Terminal Output:

```
$ git branch -a
```

```
* master
```

```
GitNewBranch
```

Switch to New Branch

Command:

```
git checkout GitNewBranch
```

Terminal Output:

```
$ git checkout GitNewBranch
```

```
Switched to branch 'GitNewBranch'
```

Create File and Add Content

Command:

```
echo "Branch Testing" > branch.txt
```

Terminal Output:

```
$ echo "Branch Testing" > branch.txt
```

Commit Changes

Command:

```
git add branch.txt
```

```
git commit -m "Added branch file"
```

Terminal Output:

```
$ git add branch.txt
```

```
$ git commit -m "Added branch file"
```

```
[GitNewBranch abc123] Added branch file
```

Check Status

Command:

```
git status
```

Terminal Output:

```
$ git status  
On branch GitNewBranch nothing to  
commit, working tree clean
```

Switch to Master

Command:

```
git checkout master
```

Terminal Output:

```
$ git checkout master  
Switched to branch 'master'
```

View Differences

Command:

```
git diff master GitNewBranch
```

Terminal Output:

```
$ git diff master GitNewBranch  
+ Branch Testing
```

View Visual Differences

Command:

```
git difftool master GitNewBranch
```

Terminal Output: \$ git difftool
master GitNewBranch Launching
P4Merge...

Merge Branch

Command:

```
git merge GitNewBranch
```

Terminal Output:

```
$ git merge GitNewBranch  
Updating abc111..abc123 Fast-  
forward
```

View Log

Command:

```
git log --oneline --graph --decorate
```

Terminal Output:

```
$ git log --oneline --graph --decorate  
* abc123 (HEAD -> master) Added branch file
```

Delete Branch

Command:

```
git branch -d GitNewBranch
```

Terminal Output:

```
$ git branch -d GitNewBranch  
Deleted branch GitNewBranch
```

Final Status

Command:

```
git status
```

Terminal Output:

```
$ git status On branch master  
nothing to commit, working tree clean
```